SOMAIYA

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K J Somaiya Institute of Technology

An Autonomous Institute Permanently Affiliated to the University of Mumbai

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Class : Second Year (SY) 2026-27 **Department : Information Technology**

Course code: ITXS37 **Subject: DSA**

Experiment No:4

Aim: Experiment No. 04 (Implementation double ended queue using Array)

Program:

```
#include <stdio.h>
#define MAX 5
int deque[MAX];
int front = -1;
int rear = -1;
int isFull() {
    return ((front == 0 && rear == MAX - 1) || (front == rear + 1));
}
int isEmpty() {
   return (front == -1);
}
void insertFront(int value) {
    if (isFull()) {
        printf("Deque Overflow! Cannot insert %d at front.\n", value);
```

```
        return;
    }
    if (front == -1) { /* Empty deque */
        front = rear = 0;
    } else if (front == 0) {
        front = MAX - 1; /* Wrap around */
    } else {
        front = front - 1;
    }
    deque[front] = value;
    printf("Inserted %d at FRONT.\n", value);
}
void insertRear(int value) {
    if (isFull()) {
        printf("Deque Overflow! Cannot insert %d at rear.\n", value);
        return;
    }
    if (front == -1) { /* Empty deque */
        front = rear = 0;
    } else if (rear == MAX - 1) {
        rear = 0; /* Wrap around */
    } else {
        rear = rear + 1;
    }
```

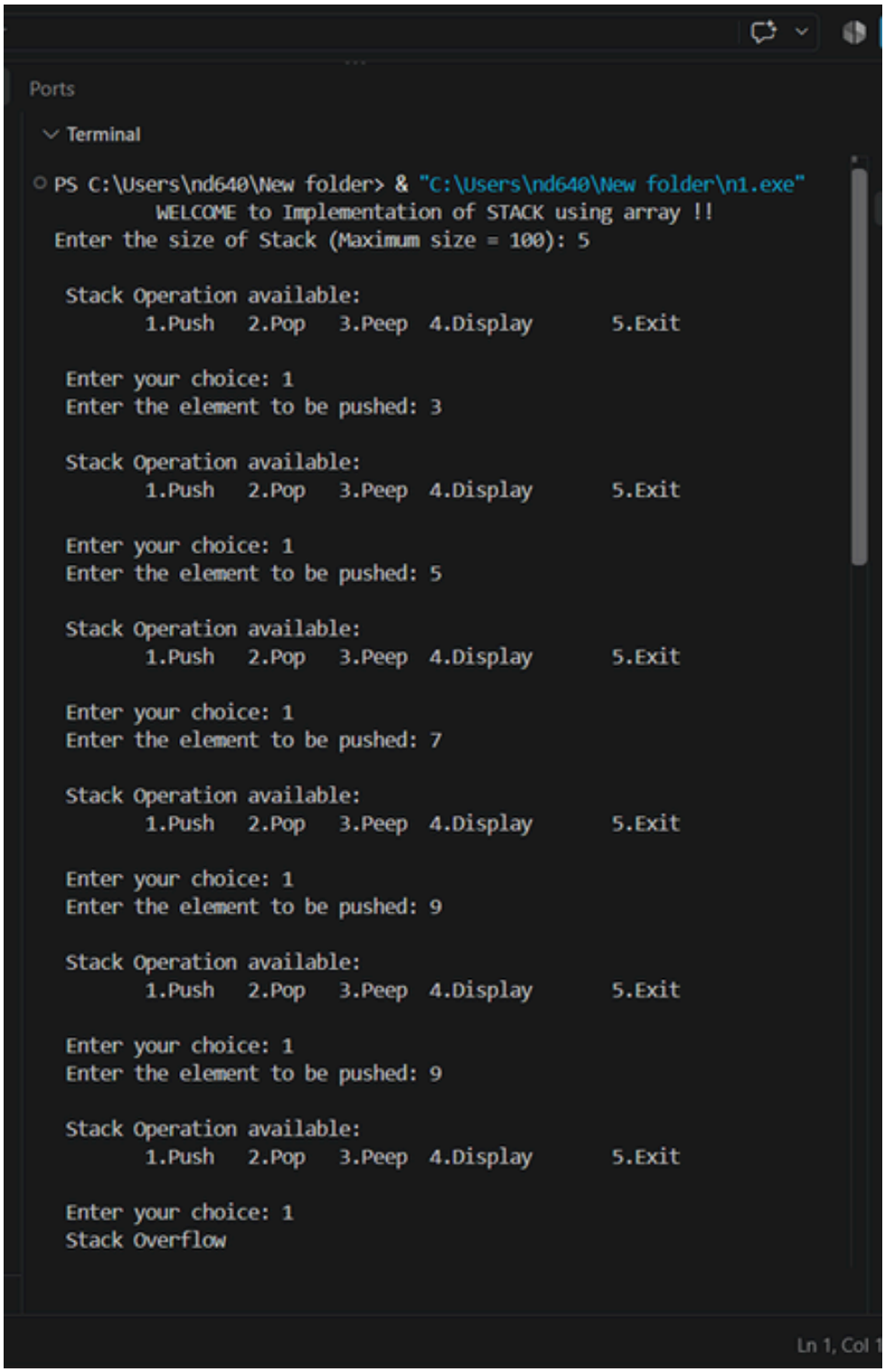
```
    deque[rear] = value;
    printf("Inserted %d at REAR.\n", value);
}
void deleteFront() {
    if (isEmpty()) {
        printf("Deque Underflow! Cannot delete from front.\n");
        return;
    }
    printf("Deleted %d from FRONT.\n", deque[front]);
    if (front == rear) { /* Only one element */
        front = rear = -1;
    } else if (front == MAX - 1) {
        front = 0; /* Wrap around */
    } else {
        front = front + 1;
    }
}
void deleteRear() {
    if (isEmpty()) {
        printf("Deque Underflow! Cannot delete from rear.\n");
        return;
    }
    printf("Deleted %d from REAR.\n", deque[rear]);
    if (front == rear) { /* Only one element */
```

```
        front = rear = -1;
    } else if (rear == 0) {
        rear = MAX - 1; /* Wrap around */
    } else {
        rear = rear - 1;
    }
}
void display() {
    if (isEmpty()) {
        printf("Deque is empty.\n");
        return;
    }
    printf("Deque elements: ");
    int i = front;
   while (1) {
        printf("%d ", deque[i]);
        if (i == rear)
            break;
        i = (i + 1) % MAX;
    }
    printf("\n");
}
int main() {
    int choice, value;
```

```
while (1) {
    printf("\n----- Deque Menu -----\n");
    printf("1. Insert at Front\n");
    printf("2. Insert at Rear\n");
    printf("3. Delete from Front\n");
    printf("4. Delete from Rear\n");
    printf("5. Display\n");
    printf("6. Exit\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);
    switch (choice) {
        case 1:
            printf("Enter value: ");
            scanf("%d", &value);
            insertFront(value);
            break;
        case 2:
            printf("Enter value: ");
            scanf("%d", &value);
            insertRear(value);
            break;
        case 3:
            deleteFront();
            break;
```

```
            case 4:
                deleteRear();
                break;
            case 5:
                display();
                break;
            case 6:
                printf("Exiting...\n");
                return 0;
            default:
                printf("Invalid choice! Try again.\n");
        }
    }
    return 0;
```

Outcome:


Ports
Terminal
PS C:\Users\nd640\New folder> & "C:\Users\nd640\New folder\n1.exe"
WELCOME to Implementation of STACK using array !!
Enter the size of Stack (Maximum size = 100): 5
Stack Operation available:
1.Push 2.Pop 3.Peep 4.Display 5.Exit
Enter your choice: 1
Enter the element to be pushed: 3
Stack Operation available:
1.Push 2.Pop 3.Peep 4.Display 5.Exit
Enter your choice: 1
Enter the element to be pushed: 5
Stack Operation available:
1.Push 2.Pop 3.Peep 4.Display 5.Exit
Enter your choice: 1
Enter the element to be pushed: 7
Stack Operation available:
1.Push 2.Pop 3.Peep 4.Display 5.Exit
Enter your choice: 1
Enter the element to be pushed: 9
Stack Operation available:
1.Push 2.Pop 3.Peep 4.Display 5.Exit
Enter your choice: 1
Enter the element to be pushed: 9
Stack Operation available:
1.Push 2.Pop 3.Peep 4.Display 5.Exit
Enter your choice: 1
Stack Overflow
Ln 1, Col 1

Conclusion: SUCESSFULLY EXCUTED THE DEQUEUE USING STRING